

Slow Learning

Christiano, Eichenbaum, Johannsen (Finance and Economics Discussion Series, Fed 2026)

Lawrence J. Christiano, Martin Eichenbaum, and Benjamin K. Johannsen

Presented by: Kamil Rogozinski

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Motivation and contribution

Learning dynamics (equation 1)

$$\theta_t = \theta_{t-1} + \gamma_t [M(\theta_{t-1}, \gamma_t) - \theta_{t-1}], \quad t\gamma_t \rightarrow 1. \quad (1)$$

θ_t is the deviation of agents' estimated belief parameters from the REE; $M(\theta_{t-1}, \gamma_t)$ is the value implied by equilibrium outcomes; and γ_t is the decreasing learning gain.

$$\text{Current belief} + \text{gain} \times (\text{equilibrium-implied belief} - \text{current belief})$$

● Literature: convergence to rational expectations equilibria

- Marcet and Sargent (1989a,b), Woodford (1990), and Evans and Honkapohja (2000, 2001) study whether learning converges to an REE.
- For the local belief map $A = D_1 M(0, 0)$, define $b = \max_i \operatorname{Re} \lambda_i(A)$. Locally, the Evans–Honkapohja condition is $b < 1 \Rightarrow \theta_t \rightarrow 0$.

● This paper: analytical speed of convergence

- For nonlinear, multivariate, decreasing-gain systems, the paper derives $\|\theta_t\| \simeq t^{b-1}$ analytically rather than estimating the rate numerically.
- The same statistic b therefore determines both local convergence and its speed.

● Policy analysis

- In calibrated NK models—especially at the ZLB—convergence can be too slow for the REE to discipline fiscal and monetary policy analysis.

Evans–Honkapohja convergence condition

$$\theta_t = \theta_{t-1} + \gamma_t [M(\theta_{t-1}, \gamma_t) - \theta_{t-1}], \quad M(0, 0) = 0, \quad t\gamma_t \rightarrow 1.$$

The associated local differential equation is

$$\dot{\theta} = M(\theta, 0) - \theta, \quad D\dot{\theta}(0) = A - I, \quad A = D_1 M(0, 0).$$

Let

$$b = \max_i \operatorname{Re} \lambda_i(A).$$

Then

$$\max_i \operatorname{Re} \lambda_i(A - I) = b - 1.$$

$$b < 1 \implies \exists U \ni 0 : \theta_0 \in U \implies \|\theta_t\| \rightarrow 0.$$

The classical result answers whether beliefs converge, not how quickly.

Analytical rate of convergence

Linearization around the fixed point gives

$$\theta_t \approx [I + \gamma_t(A - I)]\theta_{t-1}.$$

For an eigenmode associated with eigenvalue λ of A ,

$$\begin{aligned}\theta_t^{(\lambda)} &\propto \prod_{j=1}^t [1 + \gamma_j(\lambda - 1)] \\ &\approx \exp\left((\lambda - 1) \sum_{j=1}^t \gamma_j\right) \approx \exp((\lambda - 1) \log t) = t^{\lambda-1}.\end{aligned}$$

Since b is the largest real part of the eigenvalues,

$$\boxed{\|\theta_t\| \simeq t^{b-1}} \quad \text{for generic nearby initial beliefs.}$$

More precisely, for every $0 < \delta < 1 - b$,

$$\frac{\|\theta_t\|}{t^{b-1+\delta}} \rightarrow 0, \quad \frac{\|\theta_t\|}{t^{b-1-\delta}} \rightarrow \infty.$$

Horizon illustration

Define $T_{2/3}$ as the time required to eliminate two-thirds of the initial belief gap:

$$\frac{\|m_{T_{2/3}}\|}{\|m_0\|} = \frac{1}{3}, \quad T_{2/3} = \left(\frac{1}{3}\right)^{1/(b-1)} = 3^{1/(1-b)}.$$

$$b = 0.50$$

$$T_{2/3} = 3^{1/(1-0.50)} = 3^2 = 9$$

9 quarters

$$b = 0.90$$

$$T_{2/3} = 3^{1/(1-0.90)} = 3^{10} = 59,049$$

approximately 60,000 quarters

Proposition 1 and exceptional paths

Proposition 1

Let U be a neighborhood in which beliefs converge to zero.

- Uncountably many $\theta_0 \in U$ generate $\theta_t \simeq t^{b-1}$.
- For every $\theta_0 \in U$ and every $\delta > 0$,

$$\frac{\|\theta_t\|}{t^{b-1+\delta}} \rightarrow 0.$$

- The dominant rate is generic, but need not apply to every initial condition.

Two-dimensional example

$$\begin{aligned}\theta_t &= \theta_{t-1} + \gamma_t(M - I)\theta_{t-1}, \\ M &= \begin{bmatrix} b & 0 \\ 0 & a \end{bmatrix}, \quad a < b < 1.\end{aligned}$$

If $\theta_0 = [0 \ 1]'$, then the b -eigenmode is absent and

$$\theta_t = \begin{bmatrix} 0 \\ \prod_{j=1}^t [1 + \gamma_j(a-1)] \end{bmatrix}, \quad \|\theta_t\| \simeq t^{a-1}.$$

For any $\varepsilon \neq 0$,

$$\theta_0 = \begin{bmatrix} \varepsilon \\ 1 \end{bmatrix} \implies \theta_t \simeq t^{b-1};$$

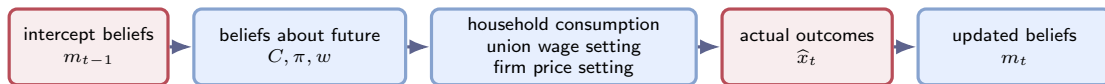
the initially tiny slow component eventually dominates.

The dominant rate is generic; special initial beliefs can converge faster by suppressing its eigenspace.

Constructing H in the NK model

Agents know the REE slope matrix Ω but estimate the intercept vector:

$$\hat{x}_{t+j} = \mu + \Omega \hat{a}_{t+j-1} + \nu_{t+j}, \quad \hat{x}_t = (\hat{C}_t, \hat{\pi}_t, \hat{w}_t)', \quad \mu = m_{t-1}.$$



Market clearing and the policy rule solve the model conditional on m_{t-1} . The resulting equilibrium map is

$$\hat{x}_t - \Omega \hat{a}_{t-1} = H m_{t-1}.$$

Agents update the perceived intercept using the equilibrium forecast error:

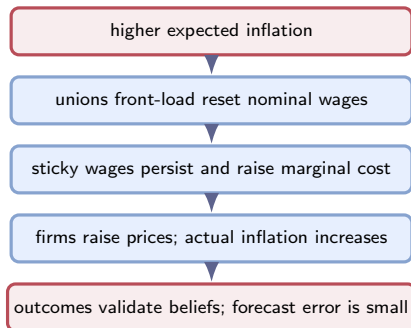
$$\begin{aligned} m_t &= m_{t-1} + \gamma_t [\hat{x}_t - \Omega \hat{a}_{t-1} - m_{t-1}] \\ &= m_{t-1} + \gamma_t (H - I) m_{t-1}. \end{aligned}$$

H is constructed from the NK model. If $Hq = bq$ with $b \approx 1$, outcomes reproduce the belief distortion in direction q , so beliefs are highly self-confirming and learning is slow.

Wage versus price rigidity

	b	$T_{2/3}$	$\hat{T}_{2/3}$
Sticky prices and wages	0.84	881	2,343
Sticky prices, flexible wages	0.26	5	6
Sticky wages, flexible prices	0.81	373	980

Removing wage rigidity collapses b from 0.84 to 0.26. Removing price rigidity leaves b near the benchmark value.



Wage rigidity—much more than price rigidity—makes beliefs self-fulfilling in the calibrated simple NK model.

Monetary policy and learning

Taylor-rule specification	b	$T_{2/3}$	$\hat{T}_{2/3}$
Benchmark: $\alpha_\pi = 1.5$, $\alpha_Y = 0.25$	0.84	881	2,343
Higher $\alpha_\pi = 2.5$	0.55	12	24
Lower $\alpha_\pi = 1.01$	0.97	1.5×10^{14}	$> 10^6$
Lower $\alpha_Y = 0.125$	0.71	42	101

Inflation response

A larger α_π produces a stronger interest-rate response to inflation. Consumption, wages, marginal cost, and inflation move closer to their REE values, creating a larger forecast error and faster updating.

Output response

When expected inflation is high and output is low, the output term offsets part of the rate increase. A smaller α_Y reduces this offset and acts like a stronger inflation response.



Medium-scale DSGE results

The CEE-style model adds

habit formation

capital and utilization

investment adjustment costs

interest-rate smoothing

Model variant	b	$T_{2/3}$	$\hat{T}_{2/3}$
CEE benchmark	0.89	33.1k	89.8k
Sticky prices, flexible wages	0.64	22	48
Sticky wages, flexible prices	0.89	15.6k	42.1k
No interest-rate smoothing	0.88	12.2k	33.0k
No habit formation	0.90	37.3k	101k
No investment adjustment costs	0.90	44.3k	120k
Higher $\alpha_\pi = 2.5$	0.70	41	98
Lower $\alpha_Y = 0.125$	0.81	376	990

Nominal rigidity

Flexible wages reduce the analytical horizon from 33,100 quarters to 22; flexible prices alone leave learning extremely slow.

Additional real frictions

Removing smoothing, habits, or investment adjustment costs does not materially change the conclusion. Wage rigidity and monetary policy remain central.

Policy implications

- **E-stability is not sufficient for policy relevance.** A locally learnable REE can be approached too slowly to describe the policy regime under study.
- **Fiscal policy at the ZLB is much weaker under learning.** The REE government-purchases multiplier is 3.95, while under learning it remains close to one because expected inflation and the real rate respond little.
- **Forward guidance is much weaker under learning.** Partially backward-looking expectations barely react to a future rate commitment; the paper finds no forward-guidance puzzle.
- **Monetary policy affects the speed of learning.** A stronger inflation response makes inflation beliefs less self-confirming and lowers b ; at the ZLB this stabilizing feedback is absent.
- **Policy evaluation should be finite-horizon.** Report the model-implied learning horizon and compare it with the expected duration of the policy episode rather than relying only on $b < 1$.

Conclusion

- **Theoretical result:** the dominant eigenvalue b that determines E-stability also determines the generic analytical power rate, $\|\theta_t\| \simeq t^{b-1}$.
- **Methodological contribution:** Proposition 1 applies to a broad class of nonlinear, multivariate, decreasing-gain systems and distinguishes generic from exceptional initial beliefs.
- **Simple NK nominal rigidities:** Table 1 shows that sticky wages drive slow learning— $b = 0.81$ with flexible prices, versus $b = 0.26$ with flexible wages.
- **Simple NK monetary policy:** raising α_π from 1.5 to 2.5 lowers b from 0.84 to 0.55, whereas $\alpha_\pi = 1.01$ makes the convergence horizon effectively unbounded.
- **Medium-scale DSGE model:** Table 2 reports extremely slow benchmark convergence, while flexible wages or a stronger inflation response sharply accelerate learning.
- **ZLB and policy:** without an active Taylor-rule response, beliefs become highly self-confirming, so REE-based fiscal multipliers and forward-guidance effects can be seriously overstated.

A rational expectations equilibrium can be learnable in the limit and still be economically irrelevant over the horizon that matters.

Appendix: Households and firms

Household problem

Household h chooses consumption and real bond holdings:

$$\max \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\log C_{h,t+j} - \frac{\chi}{2} \int_0^1 N_{i,h,t+j}^2 di \right].$$

The period budget constraint is

$$C_{h,t} + \frac{b_{h,t}}{R_t} \leq \frac{b_{h,t-1}}{\pi_t} + \int_0^1 (1 + \tau_w) w_{i,t} N_{i,h,t} di - \tau_t.$$

Variables: $C_{h,t}$ consumption; $b_{h,t}$ real bonds; R_t gross nominal interest rate; $\pi_t = P_t/P_{t-1}$ inflation; $N_{i,h,t}$ labor type i ; $w_{i,t}$ real wage; $P_{f,t}/P_t$ relative price; ε goods-demand elasticity.

Goods producers

Final-good firms demand differentiated inputs:

$$Y_{f,t} = \left(\frac{P_{f,t}}{P_t} \right)^{-\varepsilon} Y_t.$$

Intermediate firm f produces with $Y_{f,t} = N_{f,t}$.

With probability ξ_P , $P_{f,t} = P_{f,t-1}$; otherwise the firm chooses a reset price to maximize subjective expected discounted profits.

◀ Back to H construction

Appendix: Wages, policy, and aggregation

Labor unions and wage rigidity

Demand for labor type i is

$$N_{i,t} = \left(\frac{W_{i,t}}{W_t} \right)^{-\epsilon_w} N_t, \quad w_t = \frac{W_t}{P_t}.$$

With probability ξ_W , $W_{i,t} = W_{i,t-1}$; otherwise union i chooses a reset nominal wage on behalf of households.

Resource constraint

$$Y_t = C_t + G, \quad Y_t = p_t^* N_t.$$

The term p_t^* captures the output loss generated by price dispersion.

Variables: $W_{i,t}$ type-specific nominal wage; W_t aggregate nominal wage; w_t real wage; N_t aggregate labor; G government purchases; p_t^* price dispersion; α_π and α_Y Taylor-rule coefficients.

Monetary and fiscal policy

Monetary policy follows

$$R_t = \beta^{-1} + \alpha_\pi (\pi_t - 1) + \alpha_Y \frac{Y_t - \bar{Y}}{\bar{Y}}.$$

Government purchases are constant, $G_t = G$, and lump-sum taxes finance purchases and production subsidies.

Market clearing

Goods, labor, and bond markets clear. Bond-market clearing implies aggregate real bond holdings equal zero.

◀ Back to H construction

Appendix: States, beliefs, and the equilibrium map

The full aggregate vector and the variables agents learn about are

$$a_t = [C_t \quad \pi_t \quad R_t \quad Y_t \quad N_t \quad w_t \quad \tau_t \quad p_t^* \quad G_t \quad s_t]', \quad x_t = [C_t \quad \pi_t \quad w_t]'$$

Seven aggregate conditions define the mapping

$$a_t = F(x_t, a_{t-1}).$$

Agents perceive

$$\hat{x}_{t+j} = \mu + \Omega \hat{a}_{t+j-1} + \nu_{t+j}, \quad \mu = m_{t-1},$$

where Ω is fixed at its REE value and ν_{t+j} is treated as independent noise.

Conditional optimization and market clearing imply

$$\hat{x}_t - \Omega \hat{a}_{t-1} = H m_{t-1}, \quad m_t = m_{t-1} + \gamma_t (H - I) m_{t-1}.$$

Variables: τ_t taxes net of profits; s_t real marginal cost; m_t perceived intercepts; Ω known slope matrix; ν_t perceived innovation; γ_t decreasing learning gain; H equilibrium map from intercept beliefs to realized intercept-like outcomes.

$b = \max_i \operatorname{Re} \lambda_i(H)$ measures the strongest self-confirming direction.

[◀ Back to \$H\$ construction](#)